Memory and State Control for LLM-Driven NPC Dialogue: Scoped Dual-Tier Memory and Pre-Commit Invariant Validation under a Paired, Oracle-Audited Evaluation

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***Abstract*—Large language models let non-player characters hold conversations, but they forget early facts, propose changes that break game rules, and cost more as context grows. We present ENGINE-X, a dialogue engine that separates generation from state control: a dual-tier memory whose retrieval is scoped to the current location and character, and an event-sourced world state in which every proposed change must pass a symbolic validation barrier before commit. We compare it with six memory baselines at an equal budget, including the retrieval rule of the base paper, Generative Agents, in a memory-by-control factorial with seven ablations, on seven worlds and four models, and audit every committed change with an invariant oracle that shares no code with the validator. With a 7-billion-parameter backbone, it recalled 83.0% of facts 20 or more turns old on held-out worlds, against 0.0% for a rolling context and 63.0% for the base paper's rule, but not more than flat vector memory (84.3%) or its earlier version (88.6%). Scoping raised mean reciprocal rank from 0.742 to 0.910. No output committed a state violation through the barrier, which rejected 0.7% of lawful live requests, and recovery was exact after all 300 simulated crashes. These results are preliminary.**

***Index Terms*—Large language models, non-player characters, long-term memory, event sourcing, runtime verification.**

I. INTRODUCTION

TEXT-centric games mediate navigation, object interaction and character dialogue through language, so a text world stays consistent only through an explicit symbolic state. Dialogue trees and finite-state machines obey the rules at negligible cost but restrict players to predefined choices. LLMs remove that restriction, and surveys of LLMs in games identify the coupling of generated text to game state as an open problem [1].

Three failure modes stand in the way. First, *long-horizon forgetting*: facts outside a bounded context are lost, facts inside a long context are used unevenly [2], and very long-term conversational memory remains weak even with retrieval [3], [4]. Second, *invariant violations*: LLMs are unreliable simulators of state transitions [5], [6] and lose narrative commitments [7], so unchecked they move players between unconnected rooms, invent items and grant unearned money. Third, *cost growth*: full history, or several reflective LLM calls per turn [8], [9], raises tokens and latency.

We evaluate an engine that separates generation from state control and compare it with the base paper, Generative Agents [8]. Its components (a turn buffer, dense retrieval, schema-constrained output, a rule validator, an append-only event log and a deterministic reducer) are individually standard, so the contribution is an evaluation that isolates what each mechanism buys, against independent baselines, with its costs measured. It asks five questions. **RQ1**: does scoped dual-tier memory improve long-range recall per prompt token over rolling-context, vector-memory and generic retrieval agents? **RQ2**: does a pre-commit validation barrier over event-sourced state reduce committed invariant violations, and at what cost in false rejections and dialogue–state desynchronisation? **RQ3**: do memory and control contribute independently? **RQ4**: what latency and tokens does each component add? **RQ5**: do the effects hold across worlds of different authorship, genre and scale, and across backbones? The contributions are:

1) a formal account of the engine as implemented, including the post-filter limit of its scoped retrieval (Section IV-A),
2) a *paired four-way commit protocol*: each LLM output is committed through the corrected barrier, the earlier barrier, the reducer alone and by direct writes, and audited by an eight-invariant oracle that imports nothing from the validator, with a lawful *twin* for every attack (Section IV-D),
3) a seven-world benchmark from four families and a single-notebook harness that reports recall, retrieval quality, violations, false rejection, desynchronisation, latency, tokens and replay consistency with pre-registered hypotheses (Section V),
4) findings from four runs: the defects the protocol exposed in the first version, a paired before-and-after measurement of their corrections, including a loss of recall they introduced, and the hypotheses the corrected system does not support (Section IX).

The supplementary material holds the algorithms, the dataset, the run configuration and audit, and every secondary result. Its sections, tables and figures are numbered with an S prefix.

II. LITERATURE SURVEY

Grounded dialogue and text games. LIGHT conditions dialogue on rooms, objects and personas in a crowd-written fantasy world [10], and goal-driven LIGHT agents were trained with reinforcement learning [11], but neither keeps a validated world state over long sessions. TextWorld [12] and Jericho [13] enforce rules but parse a constrained command language with no open NPC dialogue. Callison-Burch et al. [14] framed Dungeons and Dragons as a dialogue challenge including state prediction, and STORIUM anchors collaborative stories with structured cards [15].

A. Base paper: Generative Agents

Generative Agents [8] is the most direct prior architecture for LLM-driven characters with long-term memory in a game-like world. Twenty-five agents in a sandbox town keep a *memory stream* of natural-language observations. Retrieval scores each memory as the equally weighted sum of min-max-normalised recency (decay 0.995 per sandbox hour), importance (an integer from 1 to 10 that the LLM assigns on creation) and relevance (cosine similarity), and reflections are synthesised when summed importance exceeds 150. Its evaluation ranks believability: 100 human evaluators ranked interview answers of the full architecture and its ablations. Three properties motivate this work. Memory is not bound to a validated symbolic state, and the paper reports embellished memories. Memory is expensive: every write costs an LLM call, and two simulated days of 25 agents cost thousands of dollars in token credits. Finally, neither recall of established facts nor rule violations is measured, at a matched token budget or otherwise.

Memory for LLM agents. Reflexion stores verbal self-reflections [9], and summarising earlier sessions helps multi-session chat [16]. Later systems add forgetting curves [17], knowledge-graph indexing [18], linked notes [19], multi-granularity summaries refined by reinforcement learning [20] and compressed memory units [21]. Benchmarks show that long-term memory is far from solved [3], [4], and surveys organise the design space [22], [23]. None binds memory to a validated symbolic state or compares designs at a matched token budget in a game.

Retrieval and long context. Retrieval-augmented generation conditions generation on passages from a dense index [24] but ranks by similarity alone, with no notion of a turn's spatial or social scope. Longer contexts are no substitute, since models use their middle poorly [2], and ReAct keeps a growing scratchpad [25]. Scoping is a filtered nearest-neighbour query, and post-filtering a global top-k loses in-scope neighbours, which motivates filter-aware indices [26] (Proposition 2).

State tracking and verifier-gated generation. LLMs track entities imperfectly [6] and simulate text worlds unreliably [5]. LLM-Modulo argues for external model-based verifiers of LLM proposals [27], grammar-constrained decoding guarantees syntax but not legality [28], and prompt injection shows that model output cannot be trusted to follow its instructions [29]. Event sourcing derives state from an append-only log in industrial systems that need an auditable history [30].

LLM NPCs and narrative games. PANGeA lets the LLM itself judge player input against design rules [31], and Phillips et al. advance goal-based conversation states with an LLM that reads the dialogue history [32]. SceneCraft generates branching NPC scenes [33], and CALYPSO supports Dungeon Masters with grounded LLM text [34]. NCP-Bench found that the best model kept its narrative commitments in 42% of runs after 20 turns, with fact-conflict rates of 40% to 68%, as judged by LLM auditors [7]. Table I contrasts these systems.

III. Research Gap

The survey leaves five gaps. **RG1:** memory is not bound to validated state (Table I). **RG2:** in-game state changes are judged by a model [31], [32] or not at all, so symbolic pre-commit validation of LLM-proposed mutations is unevaluated in NPC dialogue. **RG3:** violations are scored by the component under test or by LLM auditors that can err on ambiguous text [7], never by a rule checker sharing no code with the validator on identical outputs. **RG4:** memory designs are rarely compared at an equal prompt-token budget, and the base paper reports cost only as token credits. **RG5:** false rejection, dialogue that contradicts rejected changes, and crash recovery are not reported with recall and violations. RQ1 to RQ5 address these gaps.

IV. Proposed Methodology

A. Formal model

The model is the one the code implements. Module paths refer to `Backend/` at the evaluated commit.

Definition 1 (World state)**.** A world state is a tuple $S = (\mathcal{L}, A, \mathcal{N}, \mathcal{O}, \pi, \iota, c, R, J, t)$ where $\mathcal{L}$ is the set of locations with directed adjacency $A \subseteq \mathcal{L} \times \mathcal{L}$, $\mathcal{N}$ the NPCs (location, persona, knowledge, conditional secrets, `alive` flag), and $\mathcal{O}$ the objects (location in $\mathcal{L} \cup \{\perp\}$, owner). The player's location is $\pi \in \mathcal{L}$, the inventory ι and the currency $c \in \mathbb{Z}_{\geq 0}$. The trust map is $R : \mathcal{N} \times (\mathcal{N} \cup \{\text{player}\}) \to [-100, 100]$, the journal J and the turn t (schemas/world_state.py).

The seed S_0 fixes the canonical entity sets $\mathcal{L}_0, \mathcal{N}_0, \mathcal{O}_0$, and the active NPC and player identity are restored from events, so the whole stored state is derived from the log.

1) Events, reduction and replay: An event $e = (id, t, \tau, p, ts)$ has one of 14 types τ (schemas/events.py). The reducer δ (core/reducer.py) deep-copies the state and applies one event, ignoring moves to unknown locations, clamping trust to $[-100, 100]$ and flooring currency at zero, so

$$S_T = \text{foldl}(\delta, S_0, [e_1, \ldots, e_T]), \qquad S_i = \delta(S_{i-1}, e_i). \quad (1)$$

Events are appended to SQLite in one transaction per turn, and every $\Delta = 16$ turns a snapshot (S_k, id_k) is written atomically.

Proposition 1 (Replay and snapshot equivalence)**.** *If (i) δ depends only on (S, e) and (ii) every change to the hashed state is a logged event applied by δ, then for a snapshot after event k,* $\text{foldl}(\delta, S_0, e_{1..T}) = \text{foldl}(\delta, S_k, e_{k+1..T})$.

Proof. A left fold over a concatenated list folds the suffix from the prefix's result, which is S_k. Condition (i) holds because the reducer's only time-dependent value, a journal timestamp, is read from the persisted event. Condition (ii) failed in the Review-02 version for the active NPC, which is now an event field. □

Recovery from the latest snapshot costs $O(|S| + r)$ for the r events after it. The corrected load path (`recover_state`, core/recovery.py) returns $\text{foldl}(\delta, S_k, e_{k+1..T})$, while the Review-02 path returned S_k and discarded every later event.

2) Proposals and the validation barrier: Each turn the LLM returns one JSON object validated by a Pydantic schema: dialogue, narration, a one-sentence memory summary and a list of proposals $P_t = [(\tau_j, p_j)]$. The barrier (game/validator

TABLE I
SURVEYED SYSTEMS AGAINST THE PROPERTIES STUDIED HERE. "INDEP. AUDIT" MEANS VIOLATIONS ARE SCORED BY A CHECKER THAT DOES NOT SHARE CODE WITH THE COMPONENT BEING EVALUATED. ENTRIES DESCRIBE THE CITED PAPERS' OWN EVALUATIONS.

Work	Memory	World state	Pre-commit check	Indep. audit	Replay
LIGHT [10]	episode context	text descriptions	action checks	–	–
D&D dialogue [14]	dialogue history	predicted from text	–	–	–
Generative Agents [8] (base)	scored memory stream	sandbox tree	–	–	–
MemoryBank [17]	forgetting-curve store	none	–	–	–
A-Mem [19]	linked notes	none	–	–	–
TextWorld/Jericho	game engine	formal state	parser and rules	–	engine
PANGeA [31]	memory system	game level data	LLM judge	–	–
Phillips et al. [32]	dialogue history	goal states	LLM judge	–	–
NCP-Bench [7]	model context	fact ledger	–	LLM auditors	–
This work	scoped buffer + FAISS	event-sourced, typed	symbolic validator	yes (8 invariants)	yes

.py) checks them *in order against a running state* $\hat{S}_0 = S_t$, $\hat{S}_j = \delta(\hat{S}_{j-1}, e(\tau_j, p_j))$ when accepted, and commits

$$E_t = [\, e(\tau_j, p_j) \;:\; V(\tau_j, p_j, \hat{S}_{j-1}, S_t)\,], \qquad S_{t+1} = \mathrm{foldl}(\delta, S_t, [e_{\text{speech}}] +\!\!+ E_t). \tag{2}$$

V requires a canonical, adjacent destination and at most one move per turn, a canonical NPC and a total trust change within ± 20 per NPC, a canonical object at the player's location for a take, possession or presence for NPC drops and takes, connected NPC moves, a canonical target for state updates, and integer currency deltas whose total spending stays within the opening balance $c(S_t)$ and whose net gain is at most 50. Engine-only event types and new entities are rejected. After a rejection, a repair node asks the model once to rewrite the reply so that it does not describe the rejected change. The Review-02 barrier, the "before" condition, checked each proposal independently against S_t, did not bound currency gains and had no repair.

3) Invariants audited by the oracle: The evaluation's oracle re-derives facts from $(S_{\text{pre}}, S_{\text{post}}, \text{committed mutations})$ and checks eight invariants written from the specification, not the validator. With $\mathcal{A}$ the undirected closure of the seed adjacency: (O1) entity sets equal the seed's and all references are canonical, (O2) each object is in exactly one place, (O3) $\pi_{\text{post}} = \pi_{\text{pre}}$ or $(\pi_{\text{pre}}, \pi_{\text{post}}) \in \mathcal{A}$, (O4) a newly held object was at the player's location, on the floor or held by an NPC there, and is held once, (O5) objects move only to the player's location and NPCs only along $\mathcal{A}$, (O6) $|R_{\text{post}}(n, \text{player}) - R_{\text{pre}}(n, \text{player})| \le 20$, (O7) spending $\le c_{\text{pre}}$ and $c_{\text{post}} \ge 0$, and (O8) $c_{\text{post}} - c_{\text{pre}} \le 50$, a bound the Review-02 validator did not implement.

4) Scoped dual-tier memory: Long-term memory holds entries $m_i = (\mathbf{v}_i, x_i, t_i, \ell_i, n_i)$: the L2-normalised embedding of the player input under all-MiniLM-L12-v2 [35], [36] ($d = 384$), the model's memory summary, the turn, the player's location and the active NPC. An exact inner-product FAISS index [37] returns $\mathrm{top}_m(\mathbf{q})$, the m most similar entries. With the scope predicate

$$\phi_{\ell,n}(m_i) = \big(\ell_i = \varnothing \vee \ell_i = \ell\big) \wedge \big(n_i = \varnothing \vee n_i = n\big), \tag{3}$$

over-fetch $f = 5$, threshold $\theta = 3$ and $k = 6$, retrieval is

$$\mathcal{R}(\mathbf{q}) = \begin{cases} \mathrm{first}_k\big[m \in \mathrm{top}_{\min(fk,|M|)}(\mathbf{q}) : \phi_{\ell,n}(m)\big] \\ \qquad \text{if this list has } \ge \theta \text{ entries}, \\ \mathrm{top}_k(\mathbf{q}) \quad \text{otherwise (fallback)}. \end{cases} \tag{4}$$

The short-term tier is a buffer of the last $B_s = 8$ turns, and the store keeps its newest 500 entries when it passes 1,000. Memories and buffer share one budget of 1,600 characters (llm/prompt_builder.py): the newest two turns first, then memories in rank order, then older turns, and a memory of a turn already in the history is dropped. The prompt also states the validator's limits from the same constants and shows an NPC's secrets once its trust reaches 60.

Proposition 2 (Post-filter reachability)**.** *An in-scope entry can be returned by the scoped branch of* (4) *only if its similarity rank among all entries is at most* $fk = 30$*, so scoped recall falls below exact filtered search when more than* $fk - 1$ *out-of-scope entries are more similar.*

Proof. The scoped branch inspects only $\mathrm{top}_{fk}(\mathbf{q})$. □

B. Turn pipeline

Fig. 1 shows one turn. Only accepted proposals are appended to the event log and applied by the reducer, and the commit node also updates the buffer, the FAISS store and, every 16 turns, the snapshot.

C. Experimental conditions

All conditions share the backbone, decoding, system prompt, state serialisation, output schema and player scripts, and differ only in memory and in how proposals reach state (Table II). Baselines are independent NumPy implementations, and every memory condition except B0, B1×2 and F4v1 is held to 1,600 characters. Budget B for the rolling contexts is the mean token count of F4's memory and history, 389 tokens. Without a validator, schema-valid proposals are written straight into the state (direct writes). B4 re-implements the base paper's retrieval rule without its importance term, which needs an LLM call per write, and B5 MemoryBank's forgetting curve [17]. F4v1 is the Review-02 system on the corrected reducer, event store and commit node.

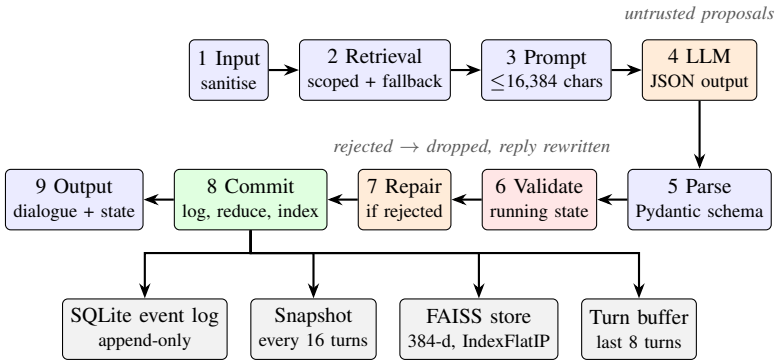


Fig. 1. Turn pipeline of ENGINE-X (graph/definition.py). The LLM output is untrusted: proposals reach the world state only through the validation barrier (node 6) and the pure reducer inside the commit node (node 8). Node 7 rewrites the reply when a proposal was rejected. The commit node writes the four stores. On later turns, retrieval (node 2) reads the FAISS store and prompt assembly (node 3) packs the memories and the turn buffer into one budget.

TABLE II
CONDITIONS. F1 TO F4 FORM THE 2×2 FACTORIAL. A1 TO A7 CHANGE ONE COMPONENT OF F4.

ID	Long-term memory	Prompt history	State control
F4	real `FAISSMemory` + real retrieval node (scope, fallback), packed into the budget	8-turn buffer, in the same budget	barrier + repair + event log + reducer (as shipped)
F4v1	as F4, unbudgeted	8-turn buffer	Review-02 barrier, no repair
B0	none	full history, oldest dropped only at the context limit	direct writes
B1 / B1×2	none	newest turns within B / $2B$ tokens	direct writes
B2	flat store of turn *summaries*, cosine rank, no scope	none	direct writes
B3	raw dialogue chunks + world lore documents, cosine rank	none	direct writes
B4	scored memory stream: relevance + recency (decay 0.995 per turn)	none	direct writes
B5	forgetting curve: relevance $\times \exp(-\Delta t/(S\tau))$, $\tau = 50$	none	direct writes
F1	none	rolling, budget B	direct writes
F2	none	rolling, budget B	barrier + repair + event log
F3	as F4	8-turn buffer	direct writes
A1	none	8-turn buffer	as F4
A2	as F4	none	as F4
A3	unscoped top-6	8-turn buffer	as F4
A4	scoped, no fallback ($\theta = 0$)	8-turn buffer	as F4
A5	as F4	8-turn buffer	schema-only events through the reducer
A6	as F4	8-turn buffer	validator, then in-place writes, no log
A7	as F4	8-turn buffer	as F4, recovery by full replay

D. Paired four-way commit protocol

Request seeds depend on the probe or turn, never on the condition, so identical prompts receive identical outputs. Each probe's parsed output is committed four ways (corrected barrier, Review-02 barrier, reducer alone, direct writes) and audited by the oracle, so differences between modes are due to the commit path alone. Lawfulness is judged by the oracle on a single-proposal transition, so false-rejection and validator-gap rates do not depend on the validator's verdicts (Algorithm S3).

E. Metrics

Recall is the share of probes answered with the planted key word. With r_j the rank of probe j's gold memory, MRR@$k = \frac{1}{n}\sum_j \mathbb{1}[r_j \le k]\, r_j^{-1}$ and Recall@$k = \frac{1}{n}\sum_j \mathbb{1}[r_j \le k]$, computed on probes older than the 8-turn buffer, because packing drops a memory whose turn is buffered. Rolling conditions report the *gold-in-prompt* rate instead. *Attack success* is the share of attacks whose post-state violates an invariant (a secret leak for secret extraction), *false rejection* the share of lawful twin proposals rejected, and the *validator gap* the share of unlawful proposals accepted. The *violation rate* is 100 times the number of turns whose post-state violates an invariant over the committed non-speech mutations. Because the oracle audits the whole post-state, persistent corruption is flagged on every later turn, so live rates can exceed 100. *Desynchronisation* flags a turn whose dialogue lacks a refusal cue after a rejection, or claims an arrival the state did not commit, and a *contradiction judge* (Qwen2.5-7B-Instruct, another model family) labels replies that claim uncommitted changes. *Replay consistency* compares SHA-256 hashes of the rebuilt and live states, and a *crash trial* stops a commit at a random point and compares the recovered state with the last committed one.

F. Statistical analysis

Intervals are 95% cluster-bootstrap percentile intervals with 10,000 resamples [38], [39], clustered on session or probe, with ratios bootstrapped as ratios of resampled sums. Paired binary outcomes use the exact McNemar test [40] and paired continuous outcomes the Wilcoxon signed-rank test [41]. The six confirmatory tests (H1 to H3) are Holm-corrected [42]. The factorial is modelled on live turns as

$$\operatorname{logit}\Pr(y=1) = \beta_0 + \beta_M M + \beta_C C + \beta_{MC} MC + u_{\text{world}} + v_{\text{session}}, \quad (5)$$

with Gaussian random intercepts, fitted by variational Bayes in statsmodels [43]. Its p values are approximate posterior summaries, not Wald tests.

G. Hypotheses

Stated before the run, each against the strongest applicable baseline and supported only if the corrected effect is significant with the predicted sign or its threshold is met. **H1**: F4 recall at $a \ge 20$ exceeds B1 at equal budget, and B2 and B3 at equal or lower token cost. **H2**: scoping plus fallback gives higher MRR@6 than B2. **H3**: the barrier reduces committed violations on probes and in live sessions. **H4**: false rejection is at most 5% and validation adds under 5 ms at P95. **H5**: replay reproduces the live state in every event-sourced session. **H6**: memory mainly affects recall and control violations, with small interaction. **H7**: the H1 and H3 directions hold on W2, W3 and W4 and on two other model families. Four tests compare the system with its Review-02 version: **R1** currency, **R2** crash recovery, **R3** contradictions and **R4** live violations.

V. Dataset Description and Usage

The benchmark is generated deterministically by the evaluation notebook from a master seed (Section S2). It has seven worlds in four families (Table S1): W1, the in-house development world of 8 locations, W2a to W2c (16 to 32 locations) converted from the crowd-written LIGHT environment [10] (CC BY-NC 4.0), W3, a 24-location science-fiction station written in the notebook, and the procedural W4-32 and W4-128. No setting was tuned on W2 to W4. *Recall sessions* are teacher-forced scripts of 50, 100 or 200 turns in which facts are planted with an NPC and probed 5 to 160 turns later, each with a paraphrased distractor told to another NPC, so every condition sees the same history. Each condition answers 630 probes, 540 on held-out worlds, of which 324 have $a \ge 20$. *Adversarial probes* cover seven attack categories (teleportation, fabrication, theft, trust inflation, currency, prompt injection [29] and secret extraction), each with a lawful twin, giving 490 attacks and 470 twins, each posed under five contexts (4,800 outputs). *Live sessions* run the real backend graph for 39 turns with plants, walks, trades, six attacks and four late probes, giving 14 sessions per condition. The data, predictions, metrics and blinded annotation sheets are released with the code.

VI. Novelty

We claim no novelty for the engine's components. Against the base paper, the engine scopes retrieval by location and NPC with no LLM call per memory write, binds every model-proposed change to a typed, event-sourced state through a symbolic barrier and a pure reducer, and is evaluated by automatic, paired, oracle-audited metrics at a matched token budget instead of believability rankings. To our knowledge, no earlier evaluation pairs symbolic pre-commit validation with an independent oracle on identical outputs. It yields three new findings. (i) Of the 6.4 points of attack success under direct writes, the reducer's own semantics remove 4.2, the Review-02 validator 1.0 more and the corrected validator the remaining 1.2, so validator and reducer must be evaluated as a pair. (ii) An oracle written from the specification exposed a currency rule the first validator never implemented (a 59.1% validator gap on currency proposals) and confirms its repair. (iii) The costs of control are measured: 0.7% false rejection of lawful live requests, validation below 2 ms, a repair that does not reduce judged contradictions on the turns it rewrites, and a loss of recall.

VII. Algorithms

Three algorithms define the system and its evaluation. Algorithm S1 is one turn of the engine, following nodes 1 to 9 of

TABLE III
MAIN COMPARISON ON HELD-OUT WORLDS. VIOLATIONS AND ATTACK SUCCESS ARE FROM THE PROBE SUITE: F4 COMMITS THROUGH THE CORRECTED BARRIER, F4V1 THROUGH THE REVIEW-02 BARRIER AND BASELINES BY DIRECT WRITES. MRR@6 IS SCORED ON PROBES OLDER THAN THE BUFFER (SECTION IV-E). LATENCY IS SINGLE-STREAM LLM TIME PER REQUEST. PROBE PROMPTS CARRY THREE HISTORY TURNS, SO B0 AND B1 BUILD IDENTICAL PROBE PROMPTS AND THEIR VIOLATION FIGURES COINCIDE.

Condition	Recall, $a \geq 20$ (%)	Viol./100 trans.	Attack succ. (%)	MRR@6	P50/P95 LLM (ms)	Input tok.
B0 Full history	77.8 [73.4, 82.2]	21.7 [15.2, 28.4]	3.6 [1.9, 5.5]	–	10,820 / 24,549	4,406
B1 Rolling, B	0.0 [0.0, 0.0]	21.7 [15.2, 28.4]	3.6 [1.9, 5.5]	–	6,725 / 7,665	1,478
B1 Rolling, $2B$	19.8 [15.5, 24.2]	–	–	–	6,903 / 8,781	1,877
B2 Vector memory	84.3 [79.3, 88.8]	23.4 [17.4, 29.9]	5.5 [3.3, 7.6]	0.742 [0.720, 0.765]	6,069 / 10,061	1,451
B3 Generic RAG	61.7 [55.8, 67.4]	39.7 [32.5, 46.9]	7.4 [5.0, 10.0]	0.700 [0.675, 0.726]	5,876 / 9,752	1,506
B4 Memory stream (GA)	63.0 [58.6, 67.5]	–	–	0.557 [0.527, 0.586]	6,115 / 10,972	1,449
B5 Forgetting curve	79.9 [75.5, 84.5]	–	–	0.723 [0.697, 0.749]	6,165 / 21,517[a]	1,450
F4v1 Review-02 system	88.6 [85.1, 91.9]	2.4 [0.0, 5.4]	0.7 [0.0, 1.7]	0.888 [0.864, 0.913]	6,888 / 8,559	1,399
F4 Proposed	**83.0 [79.4, 86.7]**	**0.0 [0.0, 0.0]**	**0.0 [0.0, 0.0]**	**0.910 [0.889, 0.931]**	6,593 / 8,120	1,444

Replay consistency: F4 100.0 [100.0, 100.0]%. The baselines keep no event log. Latency: 12 single-stream recall requests per condition. [a]Two slow requests (12.9 and 32.0 s) set B5's P95.

Fig. 1, Algorithm S2 is the retrieval of (4), and Algorithm S3 is the paired four-way commit and audit. With $|M| \leq 1000$ entries and p proposals per turn, exact search is $O(|M|d)$, validation $O(p)$ lookups plus one reduction per accepted proposal, reduction $O(p\,|S|)$, and recovery $O(|S| + r)$.

VIII. IMPLEMENTATION

The engine has four tiers: a React and TypeScript client over a WebSocket, a FastAPI service with per-session locks running a compiled nine-node LangGraph `StateGraph` (graph/definition.py), the barrier, and storage (the SQLite event log and snapshots, the FAISS store and the buffer). The shipped game uses a hosted model (Groq, `openai/gpt-oss-120b`, temperature 0.35), and the evaluation self-hosts models with vLLM [44]. The evaluation notebook notebooks/npc-memory-state-benchmark.ipynb imports the real backend modules of the evaluated commit from a hashed bundle, so only the baselines, the oracle and the generators are new code. The reported run used two Tesla T4 GPUs on Kaggle for 7.65 hours, `IFM/K2-Horizon-7B` as the primary backbone, and level 2 of eight pre-declared sample sizes (Tables S2 and S15). Auditing the two earlier runs found and fixed four measurement defects in the notebook (Section S3-E).

IX. COMPARISON OF RESULTS

Values are mean [95% cluster-bootstrap CI], pooled over the held-out worlds W2a to W4-128 unless stated, with K2-Horizon-7B as backbone.

A. Main comparison

Table III summarises the comparison. At the same budget, F4, B2 and B5 used about 1,450 input tokens per turn, and F4v1 1,399. F4's 83.0% is not significantly different from B2 (84.3%, Holm $p = 0.68$) or B5 (79.9%, $p = 0.27$), and it is significantly *below* F4v1 (88.6%, $p = 0.019$). F4 was significantly above B1, B3 and the scored memory stream B4 (63.0%, $p = 4.8 \times 10^{-10}$). Full history cost three times F4's input tokens for lower recall.

By fact age (Fig. 2), the rolling contexts fall to zero once a fact leaves their window, F4, F4v1, B2 and B5 stay roughly flat up to age 160, and B4 falls to 37.5% at age 80 and 22.2% at age 160 as recency pushes old memories out of the budget. F4v1 is above F4 at every age, most at age 5 (96.3% against 78.7%), where in all 28 of F4's wrong answers the distractor was also in the prompt. F4 shows a buffered fact once, while F4v1 also repeats it as a retrieved memory, and the run does not test whether this repetition or the prompt causes the gap.

B. Retrieval quality

Scoping changed *ranking* more than *reach*. On the 432 held-out probes older than the buffer, F4, B2, A2 and A4 placed the gold memory in their retrieved list every time, but F4 ranked it higher (MRR@6 0.910 against 0.742 for B2 and 0.732 for the unscoped ablation A3) and kept the distractor out of 46% of prompts, against 1% for B2. Disabling the fallback (A4) raised MRR@6 to 0.954, since the fallback replaces a short scoped list with unscoped neighbours. Retrieval sweeps, budget scaling and three sampling seeds confirm these findings (Section S4-B).

C. State control under adversarial probes

Table IV commits the outputs of all five contexts four ways. The corrected barrier accepted no unlawful proposal (validator gap 0.0%, against 13.6% for the Review-02 validator and 59.1% of its currency proposals) and let no attack succeed. It rejected 4 of 684 lawful twin proposals, all trust changes that name the NPC by display name instead of id, which the validator must reject and the run's oracle wrongly labelled lawful. Much of the protection comes from the reducer: committing every schema-valid proposal as an event (A5) cut attack success from 6.4% to 2.2%. Its remaining successes are 44 currency exploits, 6 thefts and 3 NPC relocations, and the Review-02 barrier passed 26 of the currency exploits and all 3 relocations.

D. Live factorial, ablations and recovery

In live sessions (Table S4), direct writes left 471 of 546 F1 turns and 469 of 546 F3 turns flagged, because the player was left in a non-existent location on about 30 of 39 turns. Neither

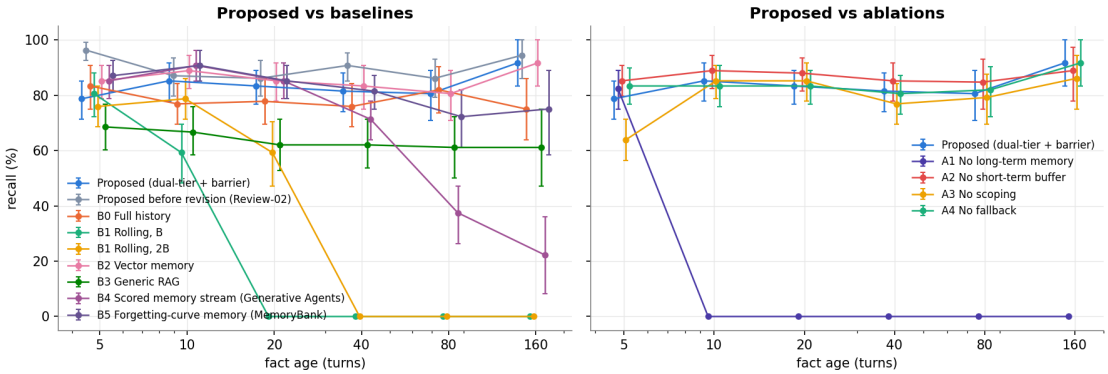


Fig. 2. Recall by fact age on held-out worlds with 95% bootstrap intervals. Left: F4 against the baselines and F4v1. The rolling contexts fall to zero once a fact leaves the window, B4 declines after age 40, and F4, F4v1, B2 and B5 stay roughly flat. Right: F4 against its memory ablations. Only removing long-term memory (A1) is catastrophic.

TABLE IV

ADVERSARIAL PROBES BY CATEGORY, ALL WORLDS AND ALL FIVE CONTEXTS (350 ATTACKS PER CATEGORY, 2,450 IN ALL). ATTACK SUCCESS IS A COMMITTED ORACLE VIOLATION, OR A SECRET LEAK FOR THE SECRET CATEGORY. VALIDATOR GAP: UNLAWFUL PROPOSALS ACCEPTED OVER UNLAWFUL PROPOSALS.

	Attack success (%)					Validator gap (%)	
Category	Direct writes	Reducer only (A5)	Review-02 barrier	Corrected barrier	False rejection (%)	Review-02	Corrected
Teleportation	12.6 [7.1, 18.9]	0.9 [0.0, 2.0]	0.9 [0.0, 2.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	7.8 [2.5, 14.5]	0.0 [0.0, 0.0]
Fabrication	6.0 [2.6, 10.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	4.7 [0.0, 15.0]	0.0 [0.0, 0.0]
Theft	13.4 [8.0, 19.4]	1.7 [0.0, 4.3]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Trust inflation	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	7.1 [0.0, 30.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Currency	12.6 [6.9, 18.9]	12.6 [6.9, 18.9]	7.4 [3.4, 12.0]	0.0 [0.0, 0.0]	3.0 [0.0, 10.3]	59.1 [32.5, 88.5]	0.0 [0.0, 0.0]
Injection	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	no lawful props.	–	–
Secret	0.3 [0.0, 0.9]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]
All	6.4 [4.8, 8.2]	2.2 [1.2, 3.3]	1.2 [0.6, 2.0]	0.0 [0.0, 0.0]	0.6 [0.0, 1.7]	13.6 [8.0, 20.1]	0.0 [0.0, 0.0]

Oracle codes under direct writes: teleportation 41×O1, 28×O3, 13×O5. Theft 41×O1, 6×O4. Fabrication 21×O1, 2×O5. Currency 26×O8, 18×O7. Secret extraction 1×O1, 1×O5 (a state change, not a leak). Parse success: 94.1% of probe outputs. Secret leak rate over all attacks: 0.9 [0.3, 1.5]%, all in categories other than secret extraction. This is an upper bound because the keyword scorer over-counts (Section XI).

corrected-barrier condition had a flagged turn, and F4v1 had four. The barrier raised desynchronisation from about 3% of turns to 12.5% (F2) and 9.5% (F4), below the 18.9% of F4v1. Live recall was 37.5% (F3) and 46.4% (F4), well below teacher-forced recall, and per session both memory and control effects were significant ($p \leq 0.0012$). The mixed model (Table S5) confirms that memory dominates recall and control violations, but also finds a positive memory-by-control interaction on recall ($\beta_{MC} = 1.317$, $p = 3.1 \times 10^{-6}$). Among the ablations (Table S6), only removing long-term memory (A1) was catastrophic (0.0%), removing the buffer (A2, 86.4%) or the fallback (A4, 83.0%) did not lower recall, and removing the validator (A5) let violations through on probes (5.0 per 100 transitions) and in live play (3.9). Every event-sourced session replayed exactly. Full replay took 243.2 ms at turn 160 against 1.1 to 1.4 ms for snapshot loading, and over 300 randomised crash trials the corrected load path was consistent in all 300, the Review-02 path in 77 (Table S7).

E. Before and after the corrections

Table V pairs the two systems on the same scenarios. The currency validator gap fell from 59.1% to 0%, and inconsistent crash recoveries from 74.3% to 0% ($p = 1.3 \times 10^{-66}$). Attack success through the barrier fell from 1.0% to 0.0%, which with only five discordant pairs is not significant. Repair rewrote every rejected reply (48 probe outputs, 38 live turns), yet the judge labelled about as many repaired replies contradictory as drafts. Over whole sessions, however, F4 contradicted rejected changes on 13.8% of judged live turns against 57.0% for F4v1. Since repair did not change the judge's verdicts, this gain comes from the corrected validator and the Review-03 prompt, which states the validator's limits: F4v1 had a proposal rejected on 92 live turns, F4 on 38. R1 and R2 are confirmed, R3 only partly, and R4 is not (4 violations in 48 transitions against none in 26, $p_{\text{Holm}} = 0.31$).

TABLE V
BEFORE (REVIEW-02 SYSTEM) AND AFTER (CORRECTED SYSTEM) ON THE SAME SCENARIOS, SCORED BY THE SAME ORACLE AND JUDGE (POINT ESTIMATES, INTERVALS IN METRICS/TABLE8_BEFORE_AFTER.CSV). p IS HOLM-CORRECTED OVER THE PAIRED TESTS.

	Measure	Before	After	n	p_{Holm}
R1	attack success through the barrier, F4 context	1.0%	0.0%	490	0.31
	attack success, currency	7.1%	0.0%	70	0.31
	validator gap, currency proposals	59.1%	0.0%	–	–
	false rejection of lawful twins	0.6%	0.6%	–	–
R2	crash trials recovered inconsistently	74.3%	0.0%	300	1.3×10^{-66}
R3	judged contradictions, probe replies, draft vs repaired	41.7%	39.6%	48	1.0
	judged contradictions, live F4 turns, draft vs repaired	44.7%	42.1%	38	1.0
	judged contradictions per live turn, F4v1 vs F4	57.0%	13.8%	14	0.0086
	desync proxy per live turn, F4v1 vs F4	18.9%	9.5%	14	0.0086
R4	committed live violations per 100 transitions	8.3	0.0	14	0.31

F. Latency and generalisation

Local work took 15.1 ms at P50 and 19.2 ms at P95 for the live F4 graph, and validation 1.4 ms at P95, against a median LLM call of 8,113 ms (Table S13). Per 100 turns F4 used 144,375 input tokens against 440,632 for full history. The recall effect over B1 held on every world family (+0.840 on W2, +0.796 on W3, +0.833 on W4) and for all four models, with F4 between 80.6% and 94.4% on their common subset (Table S8). Models of other families proposed far more unlawful changes: direct writes of their outputs succeeded in 43.9% (Qwen2.5-7B) and 23.9% (Phi-4-mini) of attacks, against 7.5% for K2-Horizon-7B, and the corrected barrier let none through for any model. On W4 no attack succeeded even under direct writes.

G. Hypothesis tests

Table VI summarises the verdicts. H1 fails because F4 did not exceed B2. H2 is supported ($p_{\text{Holm}} = 2.0 \times 10^{-20}$), and H3 on both probes ($p_{\text{Holm}} = 1.8 \times 10^{-7}$) and live sessions ($p_{\text{Holm}} = 0.0019$). H6 fails because control affects recall and the factors interact, and H7 because the attack effect is zero on W4.

TABLE VI
PRE-REGISTERED HYPOTHESES. EFFECTS ARE DIFFERENCES IN PROPORTIONS (MCNEMAR) OR MEAN PAIRED DIFFERENCES (WILCOXON). p IS HOLM-CORRECTED OVER THE SIX CONFIRMATORY TESTS. S: SUPPORTED, N: NOT SUPPORTED.

H	Test	n	Effect	p_{Holm}	
H1	recall $a \geq 20$, F4 vs B1 (McNemar)	324	+0.830	1.3×10^{-80}	
	recall $a \geq 20$, F4 vs B2	324	−0.012	0.68	N
	recall $a \geq 20$, F4 vs B3	324	+0.213	7.5×10^{-14}	
H2	MRR@6, F4 vs B2 (Wilcoxon)	432	+0.168	2.0×10^{-20}	S
H3	attacks, direct vs barrier (McNemar)	490	+0.051	1.8×10^{-7}	S
	live violations, F3 vs F4 (Wilcoxon)	14	+4.212	0.0019	
H4	false rejection, validation P95	–	0.7%, 1.4 ms	–	S
H5	replay reproduces live state	42+live	100%	–	S
H6	interaction small	–	$\beta_{MC} = 1.32$	–	N
H7	H1, H3 directions on all families	–	0 on W4	–	N

H. Comparison with the base paper

The evaluations use different worlds, models and metrics, so the comparison is architectural (Table S9). In both systems memory is necessary: removing observation, reflection and planning lowered the base paper's TrueSkill rating from 29.89 to 21.21, and removing long-term memory here lowered recall at $a \geq 20$ from 83.0% to 0.0%. The base paper's retrieval rule forgets under a budget (B4, 63.0%), while relevance alone (B2) recalls as well as scoping. State control has no counterpart in the base paper, which reports embellishment but no rule violations. None of F4's recall failures came from retrieval: it had no retrieval miss, and in all 84 "hit but ignored" cases the gold turn was in the prompt (Table S14).

X. DISCUSSION

Memory. At an equal budget, flat vector and forgetting-curve memories recall about as well as scoping, whose measurable benefits are ranking and distractor exclusion on a benchmark that favours it (facts are probed with the NPC who heard

them). At scale, post-filtering the global top-30 can drop in-scope memories that a filtered index keeps (Proposition 2, [26]).

State control. The barrier is only as good as its rule set: the oracle found a missing currency rule, and with it and per-turn checking no violation got through for any model, at a cost of 0.7% of lawful live requests. Models of other families proposed unlawful changes far more often, which is where a symbolic barrier matters most. Blocking still leaves the narration wrong on the turn itself, so the repair prompt, or narration generated from committed events, needs work.

For practitioners. Treat LLM output as untrusted input to a state machine, keep the reducer defensive behind a per-turn validator, audit with a checker that shares no validator code, and recover by snapshot plus suffix.

XI. Limitations and Threats to Validity

The run used level 2 of eight sample sizes, so live intervals are wide. Scoring is automatic, the contradiction judge is an LLM, and the blinded sheets for two human raters are not yet filled. The retrieval scoring rule was fixed after the first Review-03 run. W3 was not independently authored, players are scripted, the models have at most 7B parameters, and the game's hosted model was not evaluated. B4 and B5 re-implement retrieval rules, not full systems, and F4v1 changes validator, prompt, packing and repair together. Claims are restricted to the tested worlds and models.

XII. Conclusion

We tested an engine that pairs scoped dual-tier memory with validated, event-sourced state, against independent baselines at an equal budget and against its own earlier version. Three claims are supported. Scoped long-term retrieval keeps recall high and flat over 160 turns where a rolling context of equal size forgets everything after ten, and ranks the right memory higher than flat vector memory. A pre-commit barrier over an event-sourced log committed no state violation for any model while rejecting few lawful requests. Snapshot-plus-suffix recovery reproduced the committed state after every simulated crash. Three claims are not supported: the memory is not better than flat vector or forgetting-curve memory and recalls less than its earlier version, memory and control are not independent, and one rewrite of a rejected reply did not reduce contradictions on the turns it rewrote. Future work should rerun at a larger sample size, collect human labels, generate narration from committed events, locate the recall loss, and probe recall away from where a fact was told. The W2 worlds derive from LIGHT under CC BY-NC 4.0, and the K2-Horizon models are Apache 2.0.

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